



Advanced Linear Algebra: Foundations to Frontiers

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3.2.2 Gram-Schmidt and the QR factorization



03.2.2 Gram-Schmidt and the QR factorization

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The discussion in the last unit motivates the following theorem:

Theorem 3.2.2.1. QR Decomposition Theorem. *Let $A \in \mathbb{C}^{m \times n}$ have linearly independent columns. Then there exists an orthonormal matrix Q and upper triangular matrix R such that $A = QR$, its QR decomposition. If the diagonal elements of R are taken to be real and positive, then the decomposition is unique.*

In order to prove this theorem elegantly, we will first present the Gram-Schmidt orthogonalization algorithm using FLAME notation, in the next unit.

Ponder This 3.2.2.1. What happens in the Gram-Schmidt algorithm if the columns of A are NOT linearly independent? How might one fix this? How can the Gram-Schmidt algorithm be used to identify which columns of A are linearly independent?

▼ **Solution**

If a_j is the first column such that $\{a_0, \dots, a_j\}$ are linearly dependent, then $a_j^\perp$ will equal the zero vector and the process breaks down.

When a vector with $a_j^\perp$ equal to the zero vector is encountered, the columns can be rearranged (permuted) so that that column (or those columns) come last.

Again, if $a_j^\perp = 0$ for some j , then the columns are linearly dependent since then a_j can be written as a linear combination of the previous columns.